

q -de Rham Cohomology and Habiro Rings

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Talk 1 of SLMath Mini-workshop, September 23, 2026

These notes are based on the author's opening talk at the SLMath Mini-workshop on September 23, 2026. They survey q -de Rham cohomology, Habiro rings, and their connections to topological Hochschild homology.

1 From q -analogues to q -de Rham cohomology

The study of q -analogues has roots in Euler's work on partitions and infinite products in the 1740s. Gauss studied q -binomial coefficients in his work on quadratic Gauss sums in the 1800s. Subsequent developments include Heine's theory of q -hypergeometric series in the 1840s and Jackson's q -calculus in the 1900s.

For $n \in \mathbb{Z}$, define

$$[n]_q := \frac{1 - q^n}{1 - q} \in \mathbb{Z}[q^{\pm 1}].$$

For positive n , this is the polynomial $[n]_q = 1 + q + \cdots + q^{n-1}$, so its value at $q = 1$ is n . The same notation includes $[0]_q = 0$ and $[-n]_q = -q^{-n}[n]_q$ for $n > 0$.

The same construction extends to binomial coefficients. For $0 \leq k \leq n$, the Gaussian or q -binomial coefficient is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q := \prod_{j=1}^k \frac{1 - q^{n-k+j}}{1 - q^j} = \frac{[n]_q!}{[k]_q! [n-k]_q!},$$

where $[n]_q! := \prod_{j=1}^n [j]_q$ and $[0]_q! := 1$. Although expressed as a quotient, $\begin{bmatrix} n \\ k \end{bmatrix}_q$ lies in $\mathbb{Z}[q]$. This follows inductively from the q -Pascal recurrence $\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q + q^k \begin{bmatrix} n-1 \\ k \end{bmatrix}_q$ for $0 < k < n$, with boundary values $\begin{bmatrix} n \\ 0 \end{bmatrix}_q = \begin{bmatrix} n \\ n \end{bmatrix}_q = 1$. At $q = 1$, it specializes to $\binom{n}{k}$.

These polynomials have a geometric interpretation through point counting. When q is a prime power, let $\mathbb{F}_q$ be the field with q elements and let $\text{Gr}(k, n)$ denote the Grassmannian of k -dimensional subspaces of an n -dimensional vector space. Then

$$\# \text{Gr}(k, n)(\mathbb{F}_q) = \frac{\prod_{i=0}^{k-1} (q^n - q^i)}{\prod_{i=0}^{k-1} (q^k - q^i)} = \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

Indeed, the numerator counts ordered linearly independent k -tuples in $\mathbb{F}_q^n$, and the denominator counts ordered bases of each k -dimensional subspace. In particular, for $n \geq 1$, $\# \mathbb{P}^{n-1}(\mathbb{F}_q) = \begin{bmatrix} n \\ 1 \end{bmatrix}_q = [n]_q$. Thus q -integers and q -binomial coefficients encode counts of lines and higher-dimensional subspaces over finite fields.

Besides their role in point counting, q -integers also arise as the coefficients of a deformation of the ordinary derivative. The substitution $n \mapsto [n]_q$ is realized by a difference operator associated

with the dilation $x \mapsto qx$. The Jackson q -derivative is

$$D_q f(x) := \frac{f(qx) - f(x)}{(q-1)x}.$$

For $f \in \mathbb{Z}[q^{\pm 1}][x]$, the numerator is divisible by $(q-1)x$, so D_q defines an $\mathbb{Z}[q^{\pm 1}]$ -linear map $\mathbb{Z}[q^{\pm 1}][x] \rightarrow \mathbb{Z}[q^{\pm 1}][x]$. Its action on monomials is given by $D_q(x^n) = [n]_q x^{n-1}$. We write $d_q f = D_q f dx$.

Let γ_q denote the dilation automorphism $\gamma_q(f)(x) := f(qx)$. The Jackson derivative satisfies the *twisted Leibniz rule*

$$D_q(fg) = D_q(f)g + \gamma_q(f)D_q(g).$$

Thus D_q is a γ_q -derivation: the second term contains $f(qx)$ in place of the factor $f(x)$ in the ordinary product rule. To verify the identity, write

$$f(qx)g(qx) - f(x)g(x) = (f(qx) - f(x))g(x) + f(qx)(g(qx) - g(x))$$

and divide by $(q-1)x$.

On monomials $f = x^m$ and $g = x^n$, with $m, n \geq 1$, the rule becomes

$$D_q(x^m)x^n + (qx)^m D_q(x^n) = ([m]_q + q^m[n]_q)x^{m+n-1} = [m+n]_q x^{m+n-1} = D_q(x^{m+n}),$$

Thus the twist records the identity $[m+n]_q = [m]_q + q^m[n]_q$, which replaces the usual additivity of integers. By linearity in each factor, the monomial calculation also proves the product rule for arbitrary polynomials. For example, $D_q(x^2) = x + qx = (1+q)x$. At $q = 1$, this recovers the usual Leibniz rule.

Computation for $\mathbb{A}_{\mathbb{Z}}^1$

The effect of this deformation on cohomology can be computed explicitly on the affine line. Over $\mathbb{Z}$, the ordinary de Rham complex already has nontrivial torsion in degree one. For $X = \mathbb{A}_{\mathbb{Z}}^1 = \text{Spec}(\mathbb{Z}[x])$, the ordinary de Rham complex is

$$\Omega_{X/\mathbb{Z}}^\bullet = \left(\mathbb{Z}[x] \xrightarrow{d} \mathbb{Z}[x] dx \right), \quad d(x^n) = nx^{n-1}dx,$$

in cohomological degrees 0 and 1. Its cohomology is

$$H^0 = \mathbb{Z}, \quad H^1 = \bigoplus_{n \geq 1} (\mathbb{Z}/n\mathbb{Z}) \cdot x^{n-1}dx.$$

Its (uncompleted) q -analogue in the coordinate x is

$$C_q^\bullet = \left(\mathbb{Z}[q^{\pm 1}][x] \xrightarrow{d_q} \mathbb{Z}[q^{\pm 1}][x] dx \right).$$

Each monomial still contributes a two-term complex, but multiplication by n is replaced by multiplication by $[n]_q$. Consequently,

$$H^0(C_q^\bullet) = \mathbb{Z}[q^{\pm 1}], \quad H^1(C_q^\bullet) = \bigoplus_{n \geq 1} \mathbb{Z}[q^{\pm 1}] / ([n]_q) \cdot x^{n-1}dx.$$

Let ζ_d be a primitive d -th root of unity, with $d > 1$. First specialize q to ζ_d via $\mathbb{Z}[q^{\pm 1}] \rightarrow \mathbb{Z}[\zeta_d]$. This gives the complex $\left(\mathbb{Z}[\zeta_d][x] \xrightarrow{d\zeta_d} \mathbb{Z}[\zeta_d][x] dx \right)$. Then rationalize by tensoring over $\mathbb{Z}$ with $\mathbb{Q}$, and denote the resulting complex by

$$C_{\zeta_d}^{\bullet} := \left(\mathbb{Q}(\zeta_d)[x] \xrightarrow{d\zeta_d} \mathbb{Q}(\zeta_d)[x] dx \right).$$

The differential sends x^n to $[n]_{\zeta_d} x^{n-1} dx$. Since $[n]_{\zeta_d} = 0$ precisely when $d \mid n$, and every nonzero $[n]_{\zeta_d}$ is invertible after rationalization, its cohomology is determined by this divisibility criterion.

Proposition 1.1. *There are identifications*

$$H^0(C_{\zeta_d}^{\bullet}) = \mathbb{Q}(\zeta_d)[x^d], \quad H^1(C_{\zeta_d}^{\bullet}) = \mathbb{Q}(\zeta_d)[x^d] \cdot x^{d-1} dx.$$

Proof. On the monomial x^n , the differential is multiplication by $[n]_{\zeta_d}$, followed by $x^n \mapsto x^{n-1} dx$. This scalar is zero exactly when $d \mid n$, and is otherwise invertible in $\mathbb{Q}(\zeta_d)$. $\square$

Writing $y = x^d$, these cohomology groups identify with $\mathbb{Q}(\zeta_d)[y]$ in degree zero and $\mathbb{Q}(\zeta_d)[y] dy$ in degree one. The identifications send y to x^d and dy to $x^{d-1} dx$. The assignment on one-forms is the chosen Cartier-type cohomological identification; the ordinary pullback along $y = x^d$ would instead send dy to $d x^{d-1} dx$. These identifications express a Cartier isomorphism phenomenon in characteristic zero: the cohomology is identified with functions and one-forms in the coordinate y .

Remark 1.2. The proposition performs three operations in order: specialize q to ζ_d , rationalize by tensoring with $\mathbb{Q}$, and then take cohomology. Rationalization commutes with cohomology because $\mathbb{Q}$ is flat over $\mathbb{Z}$, so the last two operations may be interchanged. Specialization, however, need not commute with cohomology. Thus the result can differ from first taking cohomology, then specializing and rationalizing. For example, $H^0(C_q^{\bullet}) = \mathbb{Z}[q^{\pm 1}]$ becomes $\mathbb{Z}[\zeta_d]$ after specialization and $\mathbb{Q}(\zeta_d)$ after rationalization, whereas $H^0(C_{\zeta_d}^{\bullet}) = \mathbb{Q}(\zeta_d)[x^d]$. The additional classes arise because specialization makes $d_{\zeta_d}(x^{rd}) = 0$ for every $r \geq 1$.

$(q-1)$ -completed q -de Rham cohomology

The construction for $\mathbb{A}_{\mathbb{Z}}^1$ extends directly to $\mathbb{A}_{\mathbb{Z}}^n$: apply the Jackson derivative separately in each coordinate $x_1, \dots, x_n$. These operators commute and define a q -de Rham complex over $\mathbb{Z}[q^{\pm 1}]$, without completion. To extend the construction to general smooth $\mathbb{Z}$ -schemes, $X \in \text{Sm}_{\mathbb{Z}}$, one instead works with the $(q-1)$ -completed complex. This completion allows coordinate dilations to extend along étale maps, so that the construction can be made locally using framings.

Definition 1.3. A framing of a smooth $\mathbb{Z}$ -algebra R is an étale map $\square : \mathbb{Z}[x_1, \dots, x_n] \rightarrow R$.

Fix such a framing. On the polynomial algebra, let γ_i send x_i to qx_i and fix the other coordinates. These dilations reduce to the identity modulo $q-1$, so formal étaleness allows them to extend uniquely to automorphisms of $R[[q-1]]$ that are the identity modulo $q-1$. This is the reason for working in the formal neighborhood of $q=1$. The associated Jackson q -partial derivatives give the complex

$$q\Omega_{R/\mathbb{Z}, \square}^{\bullet} = \left(R[[q-1]] \longrightarrow \Omega_{R/\mathbb{Z}}^1[[q-1]] \longrightarrow \cdots \longrightarrow \Omega_{R/\mathbb{Z}}^n[[q-1]] \right).$$

For a polynomial algebra, its differential is obtained by replacing ordinary partial derivatives with Jackson q -partial derivatives. Setting $q=1$ recovers the ordinary de Rham complex.

Example 1.4. $\mathbb{A}_{\mathbb{Z}}^1$ admits the framings given by x and by $y = x + 1$. In the first framing, the dilation sends x to qx . In the second, it sends y to qy , hence sends x to $qx + (q - 1)$. Since $dy = dx$, the two differentials can be compared on the same polynomial:

$$d_{q,x}(x^2) = (1 + q)x \, dx, \quad d_{q,y}(x^2) = ((1 + q)x + (q - 1)) \, dx.$$

Indeed, $x^2 = (y - 1)^2$, so $D_{q,y}(x^2) = [2]_q y - 2 = (1 + q)x + (q - 1)$. Thus the coordinate formulas give different differentials, although both specialize to the ordinary de Rham differential at $q = 1$.

After $(q - 1)$ -completion, these coordinate complexes admit a functorial, coordinate-independent description in the derived category. Scholze formulated this coordinate-independence problem in [1]. Bhatt–Scholze’s q -crystalline theory [5] supplies the p -complete construction; Wagner gives the global construction over $\mathbb{Z}[[q - 1]]$ [3, Appendix A].

Theorem 1.5 (Bhatt–Scholze [5]; Wagner [3]). *For a smooth $\mathbb{Z}$ -scheme X , there is a functorial object $R\Gamma_{q\text{-dR}}(X) \in \mathrm{D}(\mathbb{Z}[[q - 1]])_{q-1}^\wedge$ such that*

$$R\Gamma_{q\text{-dR}}(X) \otimes_{\mathbb{Z}[[q-1]]}^L \mathbb{Z} \simeq R\Gamma_{\mathrm{dR}}(X/\mathbb{Z}).$$

On a framed smooth affine scheme, it is computed by the $(q - 1)$ -complete coordinate q -de Rham complex.

This theorem provides a cohomology theory over the formal neighborhood of $q = 1$. The earlier $\mathbb{A}_{\mathbb{Z}}^1$ calculation also involved other roots of unity, but an arbitrary formal power series in $q - 1$ cannot be evaluated at $q = \zeta_d$: the resulting series in $\zeta_d - 1$ need not converge. A theory incorporating them therefore calls for a coefficient ring that admits evaluation at every root of unity. This leads to the Habiro ring.

2 Habiro rings and Habiro cohomology

Habiro studied this ring in his work on cyclotomic completions [6]. The Habiro ring addresses the coefficient problem above by changing the completion. Instead of using only powers of $q - 1$, it uses products that eventually vanish at each root of unity. This makes evaluation possible without requiring analytic convergence of an infinite series.

Definition 2.1. Write $(q; q)_N := \prod_{j=1}^N (1 - q^j)$ for the q -Pochhammer symbol, with $(q; q)_0 := 1$. The Habiro ring is

$$\mathcal{H} := \varprojlim_N \mathbb{Z}[q]/((q; q)_N) \simeq \varprojlim_{m,n} \mathbb{Z}[q]/((q^m - 1)^n),$$

where in the second limit m is ordered by divisibility and n by size.

For any sequence of polynomials $a_n(q) \in \mathbb{Z}[q]$, the series

$$\sum_{n \geq 0} a_n(q)(q; q)_n$$

converges to an element of $\mathcal{H}$ in its inverse-limit topology; no growth condition on the coefficients is required. Indeed, for each fixed N , every term with $n \geq N$ vanishes modulo $(q; q)_N$, so the partial sums stabilize in $\mathbb{Z}[q]/((q; q)_N)$. Conversely, every $f \in \mathcal{H}$ admits such an expansion, obtained by successively lifting its compatible residues to polynomials. The coefficients representing a given f are generally not unique.

For a root of unity ζ_d , the factor $1 - q^d$ makes $(q; q)_n$ vanish at $q = \zeta_d$ whenever $n \geq d$. Thus evaluating the series leaves a finite sum and defines a map $\text{ev}_{\zeta_d} : \mathcal{H} \rightarrow \mathbb{Z}[\zeta_d]$. There is also a Taylor expansion map $T_1 : \mathcal{H} \rightarrow \mathbb{Z}[[q - 1]]$. More generally, expansion at any root of unity ζ defines a Taylor map $T_\zeta : \mathcal{H} \rightarrow \mathbb{Z}[\zeta][[q - \zeta]]$.

Theorem 2.2 (Habiro). *All maps in the following diagram are injective:*

$$\begin{array}{ccc} & & \mathbb{Z}[[q - 1]] \\ & \nearrow T_1 & \\ \mathbb{Z}[q^{\pm 1}] & \hookrightarrow \mathcal{H} & \\ & \searrow \text{ev} & \\ & & \prod_{d \geq 1} \mathbb{Z}[\zeta_d] \end{array}$$

Moreover, $T_\zeta : \mathcal{H} \rightarrow \mathbb{Z}[\zeta][[q - \zeta]]$ is injective for every root of unity ζ [6, Theorem 5.4].

The root-of-unity values satisfy arithmetic compatibility conditions, as the following proposition illustrates.

Proposition 2.3. *For every $f \in \mathcal{H}$ and every prime p ,*

$$f(\zeta_p) \equiv f(1) \pmod{1 - \zeta_p} \quad \text{in } \mathbb{Z}[\zeta_p].$$

Proof. Write $f(q) = \sum_{n \geq 0} a_n(q)(q; q)_n$, with $a_n(q) \in \mathbb{Z}[q]$. At $q = 1$, every term with $n \geq 1$ vanishes because $(q; q)_n$ contains the factor $1 - q$. Hence $f(1) = a_0(1)$.

At $q = \zeta_p$, every term with $n \geq p$ vanishes because $(q; q)_n$ contains the factor $1 - q^p$. Therefore

$$f(\zeta_p) = a_0(\zeta_p) + \sum_{n=1}^{p-1} a_n(\zeta_p) \prod_{j=1}^n (1 - \zeta_p^j).$$

Each term in this finite sum contains the factor $1 - \zeta_p$, so $f(\zeta_p) \equiv a_0(\zeta_p) \pmod{1 - \zeta_p}$. Finally, since a_0 is a polynomial with integer coefficients and $\zeta_p \equiv 1 \pmod{1 - \zeta_p}$, we have $a_0(\zeta_p) \equiv a_0(1) = f(1)$, as required. $\square$

The compatibility extends to the full Taylor expansions.

Proposition 2.4. *Choose primitive roots ζ_m so that $\zeta_{pm} - \zeta_m$ is topologically nilpotent p -adically for every prime p and $m \geq 1$. A family $(f_m(x))_{m \geq 1} \in \prod_{m \geq 1} \mathbb{Z}[\zeta_m][[x]]$ is the family of Taylor expansions of a unique element $f \in \mathcal{H}$ at $q = \zeta_m$, with $x = q - \zeta_m$, if and only if*

$$f_m(x + \zeta_{pm} - \zeta_m) = f_{pm}(x) \quad \text{in } \mathbb{Z}[\zeta_{pm}]_p^\wedge[[x]] \quad (1)$$

for every prime p and $m \geq 1$. Thus the Taylor expansion map identifies $\mathcal{H}$ with the subring of families satisfying these conditions.

Proof. Necessity follows by changing the centre of expansion for polynomial approximants and passing to the p -adic limit. Sufficiency follows from the finite-level image criterion and its p -adic formulation in [2, Propositions 5.2 and 5.3], applied to $R = \mathbb{Z}$. Uniqueness follows from the injectivity of T_1 . $\square$

Example 2.5 (Kontsevich–Zagier strange function). The series

$$F(q) := \sum_{n \geq 0} (q; q)_n$$

defines an element of $\mathcal{H}$, obtained by taking $a_n(q) = 1$ for all $n \geq 0$. It has well-defined evaluations at all roots of unity. For example, $F(1) = 1$, $F(-1) = 3$, and $F(\zeta_3) = 5 - \zeta_3$. Moreover, it admits a formal Taylor expansion at every root of unity ζ_d , with coefficients in $\mathbb{Z}[\zeta_d]$, giving an element of $\mathbb{Z}[\zeta_d][[q - \zeta_d]]$. As a series of complex numbers, however, it converges only at roots of unity; divergence at the other points of the unit circle follows from [10, Theorem 1.2].

Habiro’s original motivation: quantum invariants

Habiro’s construction was motivated by the quantum invariants of closed, oriented integral homology 3-spheres, namely manifolds M with $H_*(M; \mathbb{Z}) \cong H_*(S^3; \mathbb{Z})$. For each root of unity ζ , the normalized $\mathfrak{sl}_2$ Witten–Reshetikhin–Turaev invariant $\text{WRT}_\zeta(M)$ is a numerical topological invariant. Habiro’s unification theorem [7] realizes this family as the evaluations of a single invariant $J_M \in \mathcal{H}$, satisfying $J_M(\zeta) = \text{WRT}_\zeta(M)$ at every root of unity, with normalization $J_{S^3} = 1$. In particular, these quantum invariants are algebraic integers: $\text{WRT}_\zeta(M) \in \mathbb{Z}[\zeta]$.

The Taylor expansion $T_1(J_M) \in \mathbb{Z}[[q - 1]]$ is the Ohtsuki series of M . Since T_1 is injective, this perturbative invariant determines J_M and hence all its WRT evaluations. Moreover, the congruences between evaluations in the Habiro ring impose compatibility conditions on quantum invariants at different roots of unity.

Habiro rings of number fields

The perturbative Chern–Simons series studied in [2] have coefficients in number fields, motivating an extension of the Habiro ring to this setting.

Let $F/\mathbb{Q}$ be a number field with ring of integers $\mathcal{O}_F$. Choose a positive integer Δ divisible by $|\text{disc}(F)|$, and set $R = \mathcal{O}_F[1/\Delta]$. Thus R allows denominators supported at the primes dividing Δ , including every prime ramified in F . It is finite étale over $\mathbb{Z}[1/\Delta]$, and hence $\text{Spec}(R) \rightarrow \text{Spec}(\mathbb{Z})$ is étale. The definition of $\mathcal{H}_R$ below already applies to $R = \mathcal{O}_F[1/\text{disc}(F)]$. For each prime $p \nmid \Delta$, the p -adic completion $R_p^\wedge$ is a finite étale $\mathbb{Z}_p$ -algebra. It therefore carries a canonical Frobenius automorphism φ_p , uniquely characterized by $\varphi_p(a) \equiv a^p \pmod{p}$. These automorphisms enter the following definition.

Use the compatible roots ζ_m chosen for (1). Here $R[\zeta_m]$ means $R \otimes_{\mathbb{Z}} \mathbb{Z}[\zeta_m]$, not a compositum inside $\mathbb{C}$.

Definition 2.6 (GSWZ, Definition 1.1). The *Habiro ring of R* , denoted $\mathcal{H}_R$, is the subring of $\prod_{m \geq 1} R[\zeta_m][[x]]$ consisting of families $(f_m(x))_{m \geq 1}$ satisfying

$$f_m(x + \zeta_{pm} - \zeta_m) = \varphi_p(f_{pm}(x)) \quad \text{in } R_p^\wedge[\zeta_{pm}][[x]]$$

for every $m \geq 1$ and prime $p \nmid \Delta$. The automorphism φ_p acts on $R_p^\wedge$ and fixes ζ_{pm} and x . Addition and multiplication are componentwise.

The gluing formula compares expansions at ζ_m and ζ_{pm} after a Frobenius twist. For $R = \mathbb{Z}[1/\Delta]$, this twist is trivial, recovering (1) for primes $p \nmid \Delta$.

Example 2.7 (Constant functions and Frobenius gluing). Let $a \in R$, and consider the family in which every component is the same *constant function* $f_m(x) = a \in R[\zeta_m][[x]]$. Since translating the variable does not change a constant function, the gluing condition becomes

$$a = f_m(x + \zeta_{pm} - \zeta_m) = \varphi_p(f_{pm}(x)) = \varphi_p(a).$$

Thus this constant family belongs to $\mathcal{H}_R$ if and only if $a = \varphi_p(a)$ in $R_p^\wedge$ for every prime $p \nmid \Delta$. For example, take $R = \mathbb{Z}[i, 1/2]$ and $a = i$. At $p = 3$, the Frobenius automorphism satisfies $\varphi_3(i) = -i$, so the family $f_m(x) = i$ does not belong to $\mathcal{H}_R$. Although each component allows coefficients in R , an arbitrary $a \in R$ need not define an element of $\mathcal{H}_R$ by taking the same constant function in every component.

The authors in [2] further construct the following regulator map.

Theorem 2.8 (GSWZ). *Assume in addition that $6 \mid \Delta$. For each $\xi \in K_3(F)$, there is an invertible $\mathcal{H}_R$ -module $\mathcal{H}_{R,\xi}$, with canonical multiplication isomorphisms*

$$\mathcal{H}_{R,\xi} \otimes_{\mathcal{H}_R} \mathcal{H}_{R,\eta} \simeq \mathcal{H}_{R,\xi+\eta}, \quad \mathcal{H}_{R,0} \simeq \mathcal{H}_R.$$

Consequently, $\xi \mapsto [\mathcal{H}_{R,\xi}]$ defines a homomorphism $K_3(F) \rightarrow \text{Pic}(\mathcal{H}_R)$.

Here $\text{Pic}(\mathcal{H}_R)$ denotes the group of isomorphism classes of line bundles on $\text{Spec}(\mathcal{H}_R)$, with tensor product as the group law.

The perturbative series studied in [2] give sections of these modules: after division by local p -adic correction factors determined by ξ , their expansions satisfy the Frobenius gluing conditions. Thus these quantum invariants naturally take values in line bundles over $\mathcal{H}_R$.

Habiro cohomology and the q -Hodge complex

Habiro rings of number fields arise as the relative-dimension-zero case of a cohomology theory over $\mathcal{H}$. To perform Habiro descent, we need the additional data of a q -Hodge filtration: the object that descends is the q -Hodge complex constructed from this filtered data. Wagner [3] establishes the following existence theorem.

Theorem 2.9 (Wagner). *Let $X/\mathbb{Z}$ be smooth of relative dimension at most d . Assume that every prime $p \leq d$ is invertible on X . Then there is a canonical Habiro cohomology object $R\Gamma_{\text{Hab}}(X) \in \mathcal{D}(\mathcal{H})$. For the étale rings $R = \mathcal{O}_F[1/\Delta]$ above, it recovers $R\Gamma_{\text{Hab}}(\text{Spec } R) \simeq \mathcal{H}_R$.*

The additional input is a *q -Hodge filtration*: a filtration on the derived q -de Rham complex deforming the ordinary Hodge filtration, with the compatibility properties required for descent. Such a filtration is not canonical in general; the hypotheses of the theorem provide a canonical choice. Wagner's construction applies more generally to animated rings equipped with this data.

For a smooth affine scheme $X = \text{Spec } R$ satisfying the theorem's hypotheses, the canonical q -Hodge filtration yields a complex $q\text{-Hdg}_{R/\mathbb{Z}}$, called the *q -Hodge complex*. Wagner's descent theorem identifies it with the derived completion of Habiro cohomology:

$$R\Gamma_{\text{Hab}}(X)_{q-1}^\wedge \simeq q\text{-Hdg}_{R/\mathbb{Z}}.$$

Thus completion at $q = 1$ recovers the q -Hodge complex, not the q -de Rham complex itself. The following example gives an explicit model in polynomial coordinates.

Example 2.10. Let $R = \mathbb{Z}[x_1, \dots, x_n]$, with its standard framing $\square$, and write $C_\square^\bullet = q\Omega_{R/\mathbb{Z}, \square}^\bullet$. The coordinate q -Hodge filtration is given in degree j by

$$(\mathrm{Fil}_{q\text{-Hdg}, \square}^i C_\square^\bullet)^j = (q-1)^{\max(i-j, 0)} \Omega_{R/\mathbb{Z}}^j[[q-1]], \quad i \geq 0.$$

Thus it combines form degree with powers of $q-1$ [4, Theorem 6.10]. Denote its associated q -Hodge complex by $q\text{-Hdg}_{R/\mathbb{Z}, \square}$. The coordinate models are

$$\begin{aligned} C_\square^\bullet &= \left(\Omega_{R/\mathbb{Z}}^\bullet[[q-1]], d_{q, \square} \right), \\ q\text{-Hdg}_{R/\mathbb{Z}, \square} &\simeq \left(\Omega_{R/\mathbb{Z}}^\bullet[[q-1]], (q-1)d_{q, \square} \right). \end{aligned}$$

The extra factor arises by rescaling degree j by $(q-1)^{-j}$ in the filtered construction. In one variable, the new differential sends x^m to $(q-1)[m]_q x^{m-1} dx = (q^m - 1)x^{m-1} dx$. It therefore vanishes at $q=1$, giving $q\text{-Hdg}_{R/\mathbb{Z}, \square}/(q-1) \simeq (\Omega_{R/\mathbb{Z}}^\bullet, 0)$, whereas $C_\square^\bullet/(q-1)$ is the ordinary de Rham complex. After inverting $q-1$, multiplication by $(q-1)^j$ in degree j identifies the two models.

The subscript $\square$ records the chosen coordinate filtration. This example illustrates the filtered construction; identifying it with the canonical q -Hodge complex above requires a comparison of filtrations, beyond coordinate independence of the underlying q -de Rham complex.

3 Even filtration and Habiro cohomology

This section assumes familiarity with the motivic filtrations of Bhatt–Morrow–Scholze [12] or the even filtration of Hahn–Raksit–Wilson [9]. We state the comparison results relevant to these notes without providing a self-contained treatment of the filtrations or their construction; the details are left to subsequent talks in the workshop.

The cohomological constructions above admit descriptions in terms of (variants of) Hochschild homology and negative cyclic homology. The classical comparison recovers the Hodge filtration from negative cyclic homology. Replacing the base $\mathbb{Z}$ by connective complex K -theory ku produces its q -deformation, while periodic K -theory and finite cyclic fixed points give a description of Habiro cohomology.

The parameter q itself has a topological meaning. The standard one-dimensional representation of S^1 , acting on $\mathbb{C}$ by $z \cdot v = zv$, determines a line bundle L over $BS^1 \simeq \mathbb{CP}^\infty$. Its class defines

$$q = [L] \in \mathrm{ku}^0(BS^1) \cong \pi_0(\mathrm{ku}^{hS^1}),$$

where ku carries the trivial S^1 -action. This is the parameter identified with the q of q -de Rham cohomology by Wagner’s comparison [4, Introduction]. Tensor powers of L represent q^n ; in particular, $[n]_q = 1 + q + \dots + q^{n-1}$ is the class of $\mathcal{O} \oplus L \oplus \dots \oplus L^{\otimes(n-1)}$ for $n \geq 1$.

Restriction to the basepoint sends $[L]$ to its rank, so it sends q to 1 and $[n]_q$ to n . The reduced class $q-1 = [L] - 1$ is therefore the natural formal parameter:

$$\mathrm{ku}^0(BS^1) \cong \mathbb{Z}[[q-1]].$$

Equivalently, this is the completion of the representation ring $R(S^1) = \mathbb{Z}[q^{\pm 1}]$ at its augmentation ideal $(q-1)$. Thus the coefficient ring used earlier already appears naturally when taking circle homotopy fixed points in complex K -theory.

Even filtration on $\mathrm{TC}^-(\mathrm{ku}_R/\mathrm{ku})$

Write $\mathrm{HC}^-(R/\mathbb{Z}) = \mathrm{HH}(R/\mathbb{Z})^{hS^1}$, equipped with the S^1 -equivariant even filtration of Hahn–Raksit–Wilson [9]. The prototypical example to keep in mind is the following.

Theorem 3.1 (Antieau; Hahn–Raksit–Wilson). *For a quasi-syntomic ring R and $i \in \mathbb{Z}$, there is a natural equivalence*

$$\mathrm{gr}_{\mathrm{ev}, hS^1}^i \mathrm{HC}^-(R/\mathbb{Z}) \simeq \mathrm{Fil}_{\mathrm{Hdg}}^i \widehat{\mathrm{dR}}_{R/\mathbb{Z}}[2i].$$

This combines Antieau’s comparison [8, Theorem 1.1] with the even filtration of [9]; the formulation above is [4, Theorem 1.1]. The right side is the i -th Hodge-filtered piece, shifted by $2i$; for a smooth algebra, it is represented by $\Omega_{R/\mathbb{Z}}^{\geq i}[2i]$.

For the q -deformed comparison, write $\mathrm{TC}^-(B/A) = \mathrm{THH}(B/A)^{hS^1}$. Wagner uses an equivariant even filtration adapted to relative THH over ku [4, Sections 2–4].¹ The input includes a lift of R to the sphere spectrum $\mathbb{S}$.

Theorem 3.2 (Wagner, Theorem 4.27). *Let R be quasi-syntomic with $2 \in R^\times$. Choose a connective spherical $\mathbb{E}_2$ -lift $\mathbb{S}_R$, that is, a connective $\mathbb{E}_2$ -ring spectrum equipped with $\mathbb{S}_R \otimes_{\mathbb{S}} H\mathbb{Z} \simeq HR$, and put $\mathrm{ku}_R = \mathrm{ku} \otimes_{\mathbb{S}} \mathbb{S}_R$. The lift induces a q -Hodge filtration and equivalences*

$$\mathrm{gr}_{\mathrm{ev}, hS^1}^i \mathrm{TC}^-(\mathrm{ku}_R/\mathrm{ku}) \simeq \mathrm{Fil}_{q\text{-Hdg}}^i q\text{-}\widehat{\mathrm{dR}}_{R/\mathbb{Z}}[2i].$$

The hat denotes completion of the q -Hodge filtration. The filtration is attached to the chosen spherical lift; the comparison does not assert that it is independent of that choice.

Bott periodicization gives a direct link with the q -Hodge complex from the preceding section. Set $\mathrm{KU} = \mathrm{ku}[\beta^{-1}]$, where $\beta \in \pi_2 \mathrm{ku}$ is the Bott element, and $\mathrm{KU}_R = \mathrm{KU} \otimes_{\mathbb{S}} \mathbb{S}_R$. With Wagner’s filtered periodicization, one obtains

$$\mathrm{gr}_{\mathrm{ev}, hS^1}^0 \mathrm{TC}^-(\mathrm{KU}_R/\mathrm{KU}) \simeq q\text{-Hdg}_{R/\mathbb{Z}}$$

[4, Section 5, opening comparison]. This recovers the completed object at $q = 1$; finite cyclic symmetries supply the additional data for Habiro descent.

Habiro cohomology and $\mathrm{TC}^{-(m)}$

Continue under the hypotheses of the preceding theorem, with the same spherical lift. For $m \geq 1$, let $C_m \subset S^1$ be the cyclic subgroup of order m , and set

$$\mathrm{TC}^{-(m)}(\mathrm{KU}_R/\mathrm{KU}) := (\mathrm{THH}(\mathrm{KU}_R/\mathrm{KU})^{C_m})^{h(S^1/C_m)}.$$

Here C_m -fixed points are genuine fixed points in Wagner’s cyclonic enhancement, followed by homotopy fixed points for the residual circle. In particular, $\mathrm{TC}^{-(1)}(\mathrm{KU}_R/\mathrm{KU}) = \mathrm{TC}^-(\mathrm{KU}_R/\mathrm{KU})$.

Write $\mathrm{Fil}_{\mathrm{ev}, S^1}^\bullet \mathrm{TC}^{-(m)}(\mathrm{KU}_R/\mathrm{KU})$ for Wagner’s filtered construction, which includes the residual homotopy fixed points. For $m \mid n$, there are compatible maps from the filtered object at n to that at m [4, Remark 5.62]. Taking the inverse limit over positive integers ordered by divisibility gives

$$R\Gamma_{\mathrm{Hab}}(\mathrm{Spec} R) \simeq \mathrm{gr}^0 \left(\varprojlim_m \mathrm{Fil}_{\mathrm{ev}, S^1}^\bullet \mathrm{TC}^{-(m)}(\mathrm{KU}_R/\mathrm{KU}) \right)$$

[4, Theorem 5.63]. Here $R\Gamma_{\mathrm{Hab}}(\mathrm{Spec} R)$ denotes the Habiro object associated to the filtration induced by $\mathbb{S}_R$, called the *Habiro–Hodge complex* in that reference.

¹Wagner uses a solid version of Pstrągowski’s perfect even filtration [11]; see [4, Sections 2–3] for its construction and Section 3.4 for the comparison of filtrations.

Remark 3.3. The construction above provides a cyclonic enhancement of the S^1 -action on $\mathrm{THH}(\mathrm{KU}_R/\mathrm{KU})$. A compatible cyclotomic enhancement would require additional structure and is not supplied by this construction. In particular, the transition maps above are not the cyclotomic restriction maps used to define classical TR [4, Section 1.2]. They are instead analogous to the Frobenius maps $F : \mathrm{TR}^{r+1} \rightarrow \mathrm{TR}^r$ in the p -typical theory, which arise from inclusions of fixed points. Thus these data alone do not define the usual TC or TR for these relative inputs, even though TC^- and $\mathrm{TC}^{-(m)}$ are defined.

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